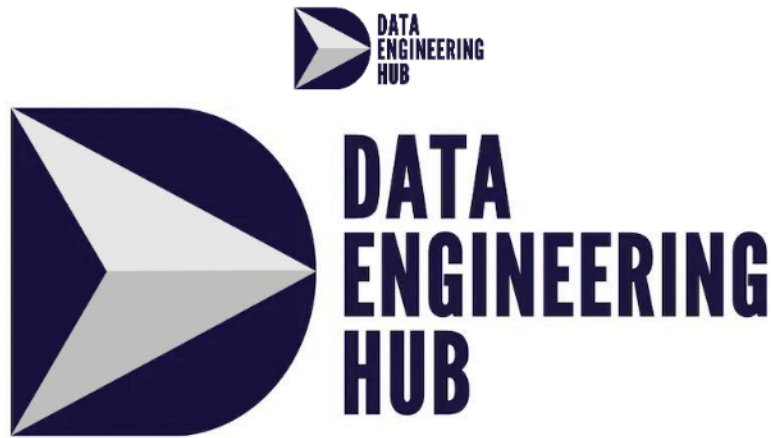




# Essential Unix Commands for Data Engineers

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# A Quick Reference Guide

## Data Engineering Hub

### Introduction

As a data engineer, you'll spend some time working with Unix/Linux command line for data processing, log analysis, and perhaps pipeline troubleshooting. This guide focuses on **practical commands** you'll actually use in your day-to-day work.

### Learning Approach

- Start with the ESSENTIAL commands (use these daily/weekly)
- Practice with real data files
- Use `chatGpt` for research and practice
- Your goal should be to internalize these over a period of 2-3 months spending 5 mins a day instead of cramming the commands!

**Remember:** *You don't need to memorize everything. Just know what's possible, then Google/research when needed!*

## ★ ESSENTIAL COMMANDS

These are your bread-and-butter commands. Learn these first – these are often used.

### Data Viewing & Analysis

| Command     | Description                         | Example Usage                         | Sample Output               |
|-------------|-------------------------------------|---------------------------------------|-----------------------------|
| <b>head</b> | View first N lines                  | <code>head -10 data.csv</code>        | First 10 rows               |
| <b>tail</b> | View last N lines /<br>monitor logs | <code>tail -10 application.log</code> | Real-time log<br>streaming  |
| <b>less</b> | Paginate through large<br>files     | <code>less big_dataset.json</code>    | Interactive file viewer     |
| <b>cat</b>  | Display entire file                 | <code>cat config.yaml</code>          | Full file content           |
| <b>wc</b>   | Count<br>lines/words/bytes          | <code>wc -l transactions.csv</code>   | 1000000<br>transactions.csv |

## File Operations

| Command     | Description            | Example Usage                                                                                                                 | Sample Output          |
|-------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>cp</b>   | Copy files/directories | <code>cp -r /data/raw<br/>/data/backup<br/>cp mission-deh-hof-ec2-<br/>setup.sh mission-deh-hof-<br/>ec2-setup_copy.sh</code> | Directory copied       |
| <b>mv</b>   | Move or rename         | <code>mv old.csv processed.csv</code>                                                                                         | File renamed           |
| <b>find</b> | Search for files       | <code>find /logs -name "*.log"<br/>-mtime -7</code>                                                                           | Files from last 7 days |

## Compression

| Command       | Description      | Example Usage                    | Sample Output                |
|---------------|------------------|----------------------------------|------------------------------|
| <b>gzip</b>   | Compress files   | <code>gzip large_file.csv</code> | Creates<br>large_file.csv.gz |
| <b>gunzip</b> | Decompress files | <code>gunzip data.csv.gz</code>  | Extracts original file       |

## Text Search & Processing

| Command     | Description              | Example Usage                                         | Sample Output          |
|-------------|--------------------------|-------------------------------------------------------|------------------------|
| <b>grep</b> | Search text patterns     | <code>grep "ERROR" spark.log</code>                   | All lines with "ERROR" |
| <b>awk</b>  | Extract/process columns  | <code>awk -F',' '{print \$1,\$5}'<br/>data.csv</code> | Columns 1 and 5        |
| <b>cut</b>  | Extract specific columns | <code>cut -d',' -f1,3 users.csv</code>                | Selected columns only  |
| <b>sed</b>  | Find and replace text    | <code>sed 's/prod/dev/g'<br/>config.txt</code>        | Text replacements      |
| <b>sort</b> | Sort lines               | <code>sort -t',' -k2 -n<br/>sales.csv</code>          | Sorted by column 2     |
| <b>uniq</b> | Remove duplicate lines   | <code>sort data.txt   uniq -c</code>                  | 3 duplicateValue       |
| <b>diff</b> | Compare two files        | <code>diff old.sql new.sql</code>                     | Shows differences      |

 **System Monitoring**

| Command   | Description          | Example Usage                       | Sample Output            |
|-----------|----------------------|-------------------------------------|--------------------------|
| <b>df</b> | Check disk space     | <code>df -h /data</code>            | 250G used of 500G        |
| <b>du</b> | Check directory size | <code>du -sh /data/warehouse</code> | 250G<br>/data/warehouse/ |

## Pipes & Redirection

| Command | Description                 | Example Usage                                     | Sample Output     |
|---------|-----------------------------|---------------------------------------------------|-------------------|
|         | Chain commands together     | <code>cat data.csv   grep "2024"   wc -l</code>   | 45230             |
| >       | Redirect output (overwrite) | <code>grep "ERROR" log.txt &gt; errors.txt</code> | Saves to new file |
| >>      | Redirect output (append)    | <code>echo "Done" &gt;&gt; status.log</code>      | Appends to file   |

## Environment & Networking

| Command       | Description               | Example Usage                                  | Sample Output            |
|---------------|---------------------------|------------------------------------------------|--------------------------|
| <b>export</b> | Set environment variables | <code>export SPARK_HOME=/opt/spark</code>      | Variable set for session |
| <b>curl</b>   | Download/API calls        | <code>curl -O https://api.com/data.json</code> | File downloaded          |
| <b>chmod</b>  | Change file permissions   | <code>chmod +x run_pipeline.sh</code>          | Makes script executable  |

## OCCASIONAL COMMANDS

You'll need these sometimes, but not daily. Good to know they exist.

| Command            | Description                 | When You'll Use It                 |
|--------------------|-----------------------------|------------------------------------|
| <code>touch</code> | Create empty file           | Creating placeholder files         |
| <code>rm</code>    | Remove files/directories    | Cleanup ( ⚠ Be careful - no undo!) |
| <code>nano</code>  | Simple text editor          | Quick file edits on server         |
| <code>vi</code>    | Advanced text editor        | When nano isn't available          |
| <code>ps</code>    | View running processes      | Finding stuck jobs                 |
| <code>kill</code>  | Terminate process           | Stopping hung processes            |
| <code>ssh</code>   | Remote server login         | Accessing data nodes               |
| <code>scp</code>   | Secure copy between servers | Transferring files across servers  |
| <code>wget</code>  | Download from web           | Getting datasets from URLs         |
| <code>which</code> | Find command location       | which python → /usr/bin/python     |

## Practice Exercises

Try these on your own to build muscle memory:

1. **Log Analysis:** `grep "ERROR" application.log | wc -l`

*Count errors in logs*

2. **Data Preview:** `head -5 data.csv | cut -d',' -f1,2`

*See first 5 rows, columns 1-2*

3. **Find Large Files:** `find /data -type f -size +100M`

*Files bigger than 100MB*

4. **Real-time Monitoring:** `tail -f pipeline.log | grep "FAILED"`

*Watch for failures*

5. **Data Validation:** `cut -d',' -f3 data.csv | sort | uniq -c`

*Count unique values in column 3*

 **Start with the ESSENTIAL commands table. Practice 5 mins daily & build confidence.**

**Good luck with your learning journey! 🎓**